

Ex:1 LINUX COMMANDS

1. **ls [options] [file]**

Displays files and directories in the current working directory.

2. **touch [options] [file]**

Creates a new empty file or updates the timestamp of an existing one.

3. **cat [options] [file]**

Prints the content of one or more files to standard output.

4. **ls -l**

Shows detailed info: permissions, links, owner, group, size, and last modified time.

5. **chmod [options] mode file**

Modifies access permissions for a file or directory.

6. **mkdir [options] directory**

Creates a new directory. Use -p to create parent directories as needed.

7. **ls -a**

Displays all files, including those starting with a dot (hidden files).

8. **ls -a | xargs -n1 xdg-open**

Lists all files including hidden ones and opens each using the default application (use with caution).

9. **nano [filename]**

Opens a simple terminal-based text editor to create or edit files.

10. **rm [options] file**

Deletes files. Use -r for directories and -f to force deletion.

11. **rmdir [options] directory**

Deletes empty directories only.

12. ps -e

Displays all currently running processes.

13. ps -e | grep [process_name] Filters the list of running processes to show only those matching the name.

14. echo "your text here"

Prints the specified text to the terminal.

15. Top

Displays live system processes, CPU/memory usage, and more.

16. free [options]

Displays total, used, and free memory in the system.

17. vmstat [options]

Shows system performance stats like processes, memory, paging, block I/O, and CPU activity.

18. ping [hostname or IP]

Sends ICMP packets to test reachability and measure response time.

19. ifconfig

Displays or configures network interfaces (deprecated in favor of `ip` command).

20. vi [filename]

Opens the `vi` editor for file editing; uses modes for navigation and editing.

Output:

ls [options] [file]

```
[User@fedora ~]$ ls
Desktop Documents Downloads Music Pictures Public Templates Videos
[User@fedora ~]$
```

touch [options] [file]

```
[User@fedora ~]$ touch newfile.txt
[User@fedora ~]$
```

cat [options] [file]

```
[User@fedora ~]$ cat .bashrc
# .bashrc

# Source global definitions
if [ -f /etc/bashrc ]; then
    . /etc/bashrc
fi

# User specific environment
if ! [[ "$PATH" =~ "$HOME/.local/bin:$HOME/bin:" ]]
then
    PATH="$HOME/.local/bin:$HOME/bin:$PATH"
fi
export PATH

# Uncomment the following line if you don't like systemctl's auto-paging feature:
# export SYSTEMD_PAGER=

# User specific aliases and functions
if [ -d ~/.bashrc.d ]; then
    for rc in ~/.bashrc.d/*; do
        if [ -f "$rc" ]; then
            . "$rc"
        fi
    done
fi

unset rc
[User@fedora ~]$
```

ls -l

```
[User@fedora ~]$ ls -l
total 0
-rw-r--r--. 1 User User  0 Sep 29 11:44 Chrome.rpm
drwxr-xr-x. 1 User User  0 Sep 29 11:37 Desktop
drwxr-xr-x. 1 User User  0 Sep 29 11:37 Documents
drwxr-xr-x. 1 User User 32 Sep 29 11:44 Downloads
drwxr-xr-x. 1 User User  0 Sep 29 11:37 Music
-rw-r--r--. 1 User User  0 Sep 29 11:48 newfile.txt
drwxr-xr-x. 1 User User 22 Sep 29 11:40 Pictures
drwxr-xr-x. 1 User User  0 Sep 29 11:37 Public
drwxr-xr-x. 1 User User  0 Sep 29 11:37 Templates
drwxr-xr-x. 1 User User  0 Sep 29 11:37 Videos
[User@fedora ~]$
```

chmod [options] mode file

```
[User@fedora ~]$ chmod 644 .bash_profile
[User@fedora ~]$
```

mkdir [options] directory

```
[User@fedora ~]$ mkdir testdir
[User@fedora ~]$ mkdir -p parentdir/childdir
```

ls -a

```
[User@fedora ~]$ ls -a
.  .bash_history  .bash_profile  .cache  .config  Documents  .local  Music  parentdir  Public  testdir
.. .bash_logout  .bashrc       Chrome.rpm Desktop  Downloads  .mozilla newfile.txt Pictures  Templates Videos
[User@fedora ~]$
```

ls -a | xargs -n1 xdg-open

```
[User@fedora ~]$ ls -a | xargs -n1 xdg-open
[User@fedora ~]$
```

nano [filename]

```
[User@fedora ~]$ nano Documents/myfile.txt
[User@fedora ~]$
```

rm [options] file

```
[User@fedora Downloads]$ rm Chrome.rpm  
[User@fedora Downloads]$
```

rmdir [options] directory

```
[User@fedora ~]$ rmdir Templates  
[User@fedora ~]$
```

ps -e

```
[User@fedora ~]$ ps -e
```

PID	TTY	TIME	CMD
1	?	00:00:01	systemd
2	?	00:00:00	kthreadd
3	?	00:00:00	rcu_gp
4	?	00:00:00	rcu_par_gp
5	?	00:00:00	slub_flushwq
6	?	00:00:00	netns
8	?	00:00:00	kworker/0:0H-events_highpri
10	?	00:00:00	mm_percpu_wq
12	?	00:00:00	rcu_tasks_kthread
13	?	00:00:00	rcu_tasks_rude_kthread
14	?	00:00:00	rcu_tasks_trace_kthread
15	?	00:00:00	ksoftirqd/0
16	?	00:00:00	rcu_preempt
17	?	00:00:00	migration/0
19	?	00:00:00	cpuhp/0
20	?	00:00:00	cpuhp/1
21	?	00:00:00	migration/1
22	?	00:00:00	ksoftirqd/1
24	?	00:00:00	kworker/1:0H-events_highpri
25	?	00:00:00	cpuhp/2
26	?	00:00:00	migration/2

ps -e | grep [process_name]

```
[User@fedora ~]$ ps -e | grep bash  
6072 pts/0 00:00:00 bash  
[User@fedora ~]$
```

echo "your text here"

```
[User@fedora ~]$ echo Hello
Hello
[User@fedora ~]$
```

top

```
[User@fedora ~]$ top

top - 12:03:02 up 31 min,  1 user,  load average: 0.24, 0.27, 0.30
Tasks: 354 total,   1 running, 353 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.2 us,  0.1 sy,  0.0 ni, 99.6 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem : 15719.7 total,  9728.1 free,  3096.2 used,  2895.4 buff/cache
MiB Swap:  8192.0 total,  8192.0 free,   0.0 used. 12271.8 avail Mem

  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  COMMAND
 3890 User        20   0 6139048 286016 152364 S   2.0   1.8   0:45.56 gnome-shell
 6054 User        20   0  773148  59876  46928 S   1.7   0.4   0:04.18 gnome-terminal-
 4907 User        20   0 4050312 591072 266988 S   0.3   3.7   1:18.87 firefox
 5293 User        20   0 3485844 603000 142212 S   0.3   3.7   0:39.98 Isolated Web Co
 5626 User        20   0 3063408 387472 121032 S   0.3   2.4   0:12.42 Isolated Web Co
 6431 root         20   0      0      0      0 I   0.3   0.0   0:00.34 kworker/u24:0-events_unbound
 7523 User        20   0 226620  4504   3488 R   0.3   0.0   0:00.03 top
    1 root         20   0 173764  18564 11332 S   0.0   0.1   0:01.52 systemd
    2 root         20   0      0      0      0 S   0.0   0.0   0:00.00 kthreadd
    3 root          0 -20      0      0      0 I   0.0   0.0   0:00.00 rcu_gp
    4 root          0 -20      0      0      0 I   0.0   0.0   0:00.00 rcu_par_gp
    5 root          0 -20      0      0      0 I   0.0   0.0   0:00.00 slub_flushwq
    6 root          0 -20      0      0      0 I   0.0   0.0   0:00.00 netns
    8 root          0 -20      0      0      0 I   0.0   0.0   0:00.00 kworker/0:0H-events_highpri
   10 root          0 -20      0      0      0 I   0.0   0.0   0:00.00 mm_percpu_wq
   12 root         20   0      0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_kthread
   13 root         20   0      0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_rude_kthread
   14 root         20   0      0      0      0 I   0.0   0.0   0:00.00 rcu_tasks_trace_kthread
   15 root         20   0      0      0      0 S   0.0   0.0   0:00.01 ksoftirqd/0
   16 root         20   0      0      0      0 I   0.0   0.0   0:00.57 rcu_preempt
   17 root          rt   0      0      0      0 S   0.0   0.0   0:00.00 migration/0
   19 root         20   0      0      0      0 S   0.0   0.0   0:00.00 cpuhp/0
   20 root         20   0      0      0      0 S   0.0   0.0   0:00.00 cpuhp/1
   21 root          rt   0      0      0      0 S   0.0   0.0   0:00.12 migration/1
```

free -h

```
[User@fedora ~]$ free -h

              total        used        free       shared    buff/cache   available
Mem:          15Gi         3.0Gi         9.5Gi          78Mi         2.8Gi         11Gi
Swap:         8.0Gi           0B         8.0Gi
```

vmstat [options]

```
[User@fedora ~]$ vmstat
procs -----memory----- ---swap-- -----io----- -system-- -----cpu-----
 r  b   swpd   free   buff  cache   si   so    bi    bo   in   cs us sy id wa st
  1  0       0 9980136  4704 2965280    0    0    74    32  151  263  1  0 97  1  0
[User@fedora ~]$
```

ping [hostname or IP]

```
[User@fedora ~]$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data.
64 bytes from 8.8.8.8: icmp_seq=1 ttl=118 time=5.59 ms
64 bytes from 8.8.8.8: icmp_seq=2 ttl=118 time=6.20 ms
64 bytes from 8.8.8.8: icmp_seq=3 ttl=118 time=5.94 ms
64 bytes from 8.8.8.8: icmp_seq=4 ttl=118 time=5.40 ms
64 bytes from 8.8.8.8: icmp_seq=5 ttl=118 time=5.61 ms
64 bytes from 8.8.8.8: icmp_seq=6 ttl=118 time=5.66 ms
^C
--- 8.8.8.8 ping statistics ---
6 packets transmitted, 6 received, 0% packet loss, time 5010ms
rtt min/avg/max/mdev = 5.400/5.731/6.197/0.262 ms
[User@fedora ~]$
```

ifconfig

```
[User@fedora ~]$ ifconfig
enp2s0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 172.16.52.107 netmask 255.255.254.0 broadcast 172.16.53.255
    inet6 fe80::7689:f37e:a232:ae47 prefixlen 64 scopeid 0x20<link>
    ether 88:ae:dd:15:ed:5b txqueuelen 1000 (Ethernet)
    RX packets 251240 bytes 295683148 (281.9 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 81081 bytes 18222321 (17.3 MiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0x10<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 42 bytes 5489 (5.3 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 42 bytes 5489 (5.3 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

[User@fedora ~]$
```

vi [filename]

```
[User@fedora ~]$ vi Documents/myfile.txt
[User@fedora ~]$
```

E325: ATTENTION

```
Found a swap file by the name "Documents/.myfile.txt.swp"
  owned by: User   dated: Mon Sep 29 12:08:51 2025
  file name: ~User/Documents/myfile.txt
  modified: YES
  user name: User   host name: fedora
  process ID: 7720
```

```
While opening file "Documents/myfile.txt"
  CANNOT BE FOUND
```

- (1) Another program may be editing the same file. If this is the case, be careful not to end up with two different instances of the same file when making changes. Quit, or continue with caution.
- (2) An edit session for this file crashed.
If this is the case, use `:recover` or `"vim -r Documents/myfile.txt"` to recover the changes (see `":help recovery"`).
If you did this already, delete the swap file `"Documents/.myfile.txt.swp"` to avoid this message.

```
"Documents/myfile.txt" [New]
```

```
Press ENTER or type command to continue
```